Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
2	*(a)*	(i)		Mean = 50 cm (1) [accept 49.8 cm] Uncertainty = $\frac{(54 - 46)}{2}$ = [±] 4 cm (1) [accept 4.0 cm] **Alternative for 2nd mark:** Uncertainty = 54 - 49.8 = 4.2 cm (1) [accept 4.0 cm or 4 cm]		2		2	2	2
		(ii)		% uncertainty = $\left(\frac{4}{50}\right) \times 100$ **(ecf on mean and uncertainty)** = ± 8% accept 8.4 % if alternative method used in (i) allow 1 or 2 sig figs		1		1	1	1
	(b)	(i)		Conservation of energy (1) accept energy can't be created or destroyed – don't accept principle of energy Energy held in / work done by compressed spring transferred to gravitational potential energy (1) Accept: $\frac{1}{2} kx^2 = mgh_{mean}$	1	1		2		
		(ii)	I	Very small uncertainty in both measurements **or** high precision **or** small resolution [when compared to uncertainty in jump height] (1) Accept calculations – 0.3% and 0.02%. Don't accept reference to accuracy Hence negligible effect on [overall] uncertainty **or** when compared to 8% (1)			2	2		2
			II	Substitution: e.g. $k = \frac{(2 \times 48.4 \times 10^{-3} \times 9.81 \times 50)}{3.20^2}$ (**ecf on mean**) Ignore powers of 10 (1) k = 4.6 N cm^{-1} **or** 460 N m^{-1} **(unit mark)** (1) accept 4.62 or 462 if 49.8 used and unit J m^{-2} Uncertainty = ± 8% × 4.64 = ± 0.4 N cm^{-1} or 40 N m^{-1} (1) (**ecf on both % uncertainty and k**) **Allow 1 or 2 sig figs or same number of sig figs as h** **Accept answers when uncertainties in mass and x have been calculated by candidates e.g. total uncertainty = 8.32%**	1	1 1		3	3	3

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Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
			III	Candidate's answer to k too low or actual value of k higher (1) h affected by air resistance or energy lost due to vibrations during take-off or toy moves slightly before release so x is not = 3.20 cm or less than 100% of the EPE is transferred to GPE (1) Don't accept only general references to sound and heat			2	2		2
	(c)			Use freeze frame technology/ camera/ i-Pad etc to measure jump height [with greater precision or more accurately] Multiple repeats <u>of h</u> [to reduce random error] Accept laser or light distance measuring system to measure distance [with greater precision or more accurately] Don't accept use of light gates or using a different spring or reduce parallax error			1	1		1
				Question 2 total	**2**	**6**	**5**	**13**	**6**	**11**

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